

V4.2 Toy Model 3B

Dynamics from Phase Rotation: Toward Schrodinger-Type Evolution

Rev A - Exploratory Mathematical Compatibility Report

Date: 2026.05.25 | Continuation after Toy Model 3A

Status: Toy Model 3B tests whether first-order quantum-style time evolution can be represented as real rotation in the hidden transverse phase plane. It does not derive the physical Hamiltonian of any real particle and does not derive the full Schrodinger equation from first principles.

1. Purpose

Toy Model 3A showed that complex phase can be represented as real rotation in the two hidden transverse temporal directions left by the 3T -> 1T projection. Toy Model 3B now tests dynamics: if the phase vector rotates uniformly as observable 1T time advances, does the compact complex equation take the Schrodinger-type form?

The target is structural compatibility. We are not deriving the real physical Hamiltonian for electrons, atoms, fields, or the Standard Model.

2. Starting point from 3A

The real transverse phase plane has the 90-degree rotation operator:

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$J^2 = -I$$

In complex shorthand, the same rotation is represented by multiplication by i .

3. Uniform hidden phase rotation

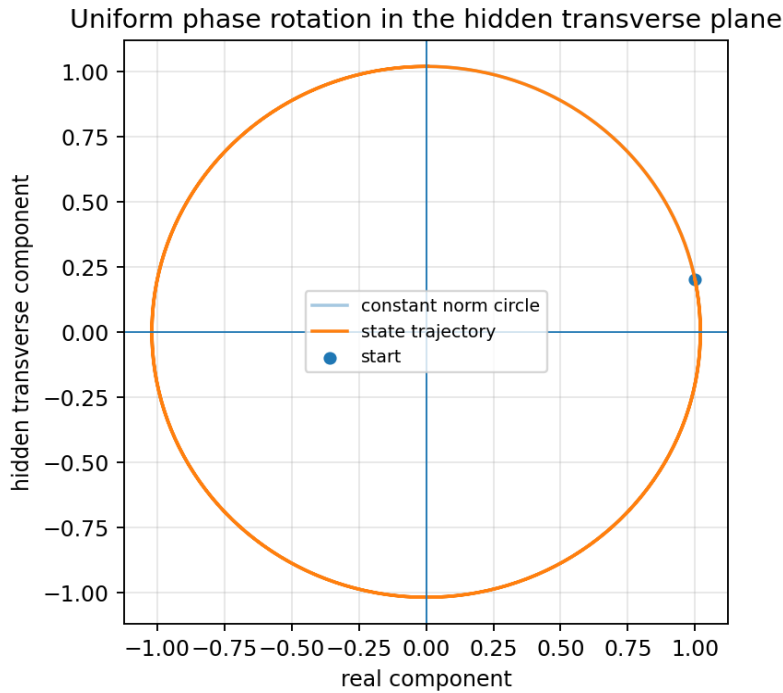
Let $v(t)$ be a real two-component phase-plane vector. Uniform phase rotation with angular frequency ω is:

$$v(t) = R(-\omega t) v(0)$$

$$R(-\omega t) = \cos(\omega t) I - \sin(\omega t) J$$

Term reminders:

- ω is angular frequency, meaning rotation rate in radians per unit time.
- t is observable 1T time.
- J is the 90-degree rotation operator in the hidden transverse phase plane.
- $R(-\omega t)$ is a real rotation matrix.



4. First-order differential equation

Differentiating the real rotation gives:

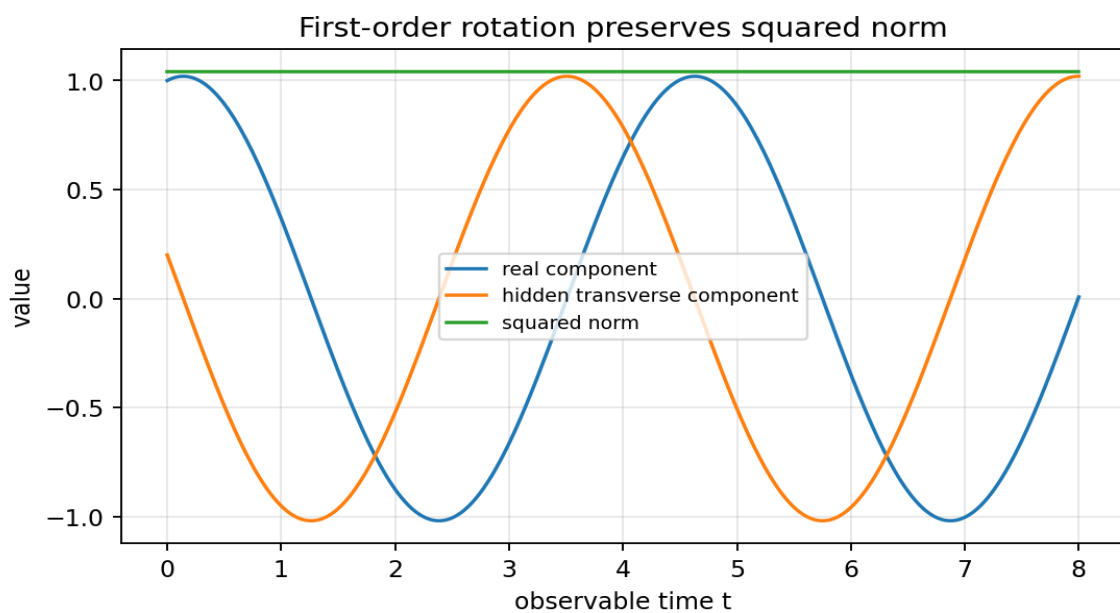
$$d \mathbf{v} / dt = - \omega \mathbf{J} \mathbf{v}$$

Plain meaning: the time derivative of the state is generated by a 90-degree rotation of the state. This is the real-plane form of phase dynamics.

Because $\mathbf{J}$ is antisymmetric, this motion preserves squared length:

$$\begin{aligned} d/dt (\mathbf{v} \cdot \mathbf{v}) &= 2 \mathbf{v} \cdot (d\mathbf{v}/dt) \\ &= -2 \omega \mathbf{v} \cdot \mathbf{J} \mathbf{v} \\ &= 0 \end{aligned}$$

The last equality holds because $\mathbf{v} \cdot \mathbf{J} \mathbf{v} = 0$ for an antisymmetric $\mathbf{J}$. Therefore the norm is preserved. Norm preservation is the real-geometric version of unitary evolution.



5. Complex shorthand and Schrodinger-type form

Use the complex shorthand from 3A:

$$\mathbf{v} = (u, w) \leftrightarrow \psi = u + i w$$

The real equation becomes:

$$d \psi / dt = - i \omega \psi$$

Multiplying both sides by i gives:

$$i \frac{d \psi}{dt} = \omega \psi$$

If we introduce $\hbar$ and define energy $E = \hbar \omega$, this becomes:

$$i \hbar \frac{d \psi}{dt} = E \psi$$

This has the same structural form as the energy-eigenstate Schrodinger equation. It says that a fixed-energy state evolves by phase rotation.

6. Multiple energy components

For multiple independent energy components, each component has its own frequency ω_k :

$$i \frac{d \psi_k}{dt} = -\omega_k \psi_k$$

In complex notation:

$$i \frac{d \psi_k}{dt} = \omega_k \psi_k$$

With $E_k = \hbar \omega_k$:

$$i \hbar \frac{d \psi_k}{dt} = E_k \psi_k$$

Stacking the components gives the diagonal Hamiltonian form:

$$i \hbar \frac{d \psi}{dt} = H \psi$$

Here H is diagonal in the chosen energy basis. This is not yet a derivation of H ; it shows how Schrodinger-type evolution arises once phase rotation rates are assigned.

7. General finite-dimensional Hamiltonian as real rotation generator

A general finite-dimensional Hermitian Hamiltonian can also be represented as a real antisymmetric generator on doubled real coordinates. Let:

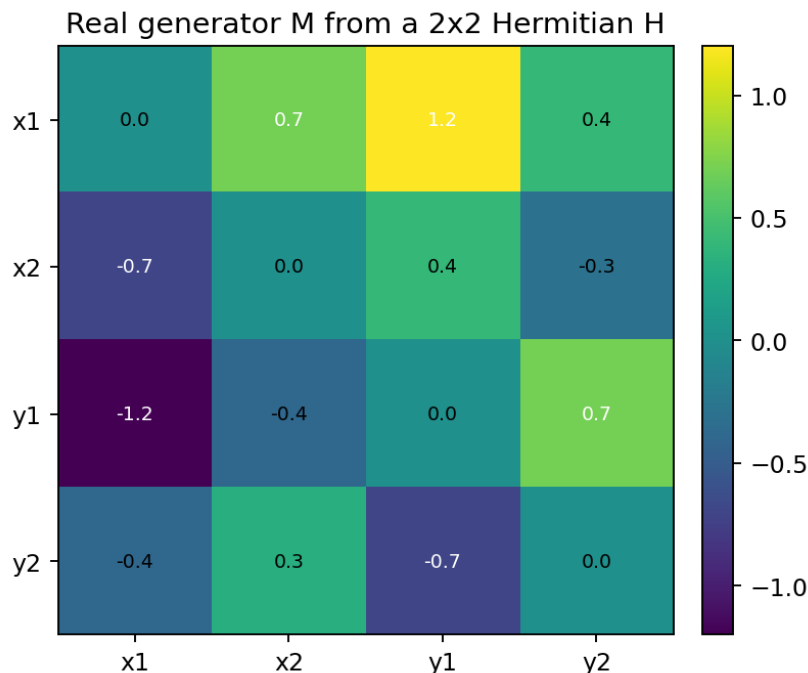
$$H = A + i B$$

$$\begin{aligned} A^T &= A & (A \text{ is real symmetric}) \\ B^T &= -B & (B \text{ is real antisymmetric}) \end{aligned}$$

For $\psi = x + i y$, using stacked real coordinates $[x_{\text{all}}; y_{\text{all}}]$, the equation $d \psi / dt = -i H \psi$ becomes:

$$\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} B & A \\ -A & B \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

The real matrix on the right is antisymmetric. Therefore it preserves the real squared norm $x \cdot x + y \cdot y$.



8. Numerical checks

Check	Result
Random samples	10000

Check	Result
Max $J^2 + I$ error	0.000e+00
Max real-rotation / complex-phase equivalence error	4.441e-16
Max finite-difference derivative error	2.902e-08
Max single-plane norm preservation error	3.553e-15
Max block generator skew-symmetry error	0.000e+00
Max block norm-derivative error	7.773e-15
Max general Hermitian real-generator skew error	0.000e+00
Max general complex/real derivative equivalence error	1.776e-15
Max general norm-derivative error	8.593e-15

9. Interpretation inside the 6D Projection Framework

Toy Model 3B supports the following limited interpretation:

```
observable 1T advancement
  -> continuous rotation in hidden transverse 2T phase plane
  -> real first-order equation  $dv/dt = -\omega J_v$ 
  -> complex shorthand  $d\psi/dt = -i \omega \psi$ 
  -> Schrodinger-type structure  $i \hbar d\psi/dt = H \psi$ 
```

This makes quantum phase evolution geometrically interpretable as hidden temporal rotation. The imaginary unit is not added as a mystery symbol; it is the compact notation for the real rotation operator J identified in 3A.

10. What Toy Model 3B achieves

- It starts from real phase-plane rotation rather than assuming complex dynamics.
- It derives the first-order real equation $dv/dt = -\omega J_v$ for uniform phase rotation.
- It shows that the complex shorthand gives $i d\psi/dt = \omega \psi$.
- It recovers the Schrodinger-type energy-eigenstate form $i \hbar d\psi/dt = E \psi$.
- It shows norm preservation as a consequence of antisymmetric real generators.
- It shows that finite-dimensional Hermitian Hamiltonians correspond to real antisymmetric generators on doubled real coordinates.

11. What Toy Model 3B does not achieve

- It does not derive the physical Hamiltonian H for any real system.
- It does not derive the full Schrodinger equation with kinetic and potential terms.
- It does not derive quantum field theory or the Standard Model.
- It does not derive the Born rule.
- It does not prove physical 3T existence.
- It does not yet produce new experimental predictions.

12. Close-out statement for 3B

Toy Model 3B successfully constructs a structural bridge from real hidden phase-plane rotation to Schrodinger-type time evolution. Uniform rotation in the hidden transverse 2T plane gives the real equation $dv/dt = -\omega J_v$. Using the complex shorthand from 3A, this becomes $i d\psi/dt = \omega \psi$, and with $E = \hbar \omega$ it takes the energy-eigenstate Schrodinger form $i \hbar d\psi/dt = E \psi$. More generally, Hermitian Hamiltonian evolution corresponds to real antisymmetric norm-preserving generators on doubled real coordinates. This is a structural compatibility result, not a derivation of the physical Hamiltonian, the full Schrodinger equation, or quantum field theory.

13. Recommended next chapter

The next logical chapter is Toy Model 3C: Recovering the Kinetic-Term Structure - Toward Spatial Wave Dynamics.

1. Introduce spatial dependence $\psi(x,t)$.
2. Test how phase gradients relate to momentum.
3. Check whether a second spatial derivative term can arise from local phase-coupled propagation.

4. Recover only the structural form of the free-particle Schrodinger equation first.
5. Keep the guardrail: do not claim a full derivation until mass, momentum, and kinetic energy are connected consistently.

14. Rebind prompt for continuation

We are resuming after Toy Model 3B.

3B tested dynamics from hidden phase rotation.

From 3A:

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$J^2 = -I$$

Uniform hidden phase rotation:

$$v(t) = R(-\omega t) v(0)$$

$$R(-\omega t) = \cos(\omega t) I - \sin(\omega t) J$$

Differential equation:

$$dv/dt = -\omega J v$$

Complex shorthand:

$$v = (u, w) \leftrightarrow \psi = u + i w$$

Then:

$$d\psi/dt = -i \omega \psi$$

$$i d\psi/dt = \omega \psi$$

With $E = \hbar \omega$:

$$i \hbar d\psi/dt = E \psi$$

Result:

Schrodinger-type time evolution can be represented as hidden transverse temporal phase rotation.

Guardrail:

3B does not derive the physical Hamiltonian H , the full Schrodinger equation, the Born rule, QFT, or physical 3T existence.

Next recommended chapter:

Toy Model 3C - Recovering the Kinetic-Term Structure: toward spatial wave dynamics.